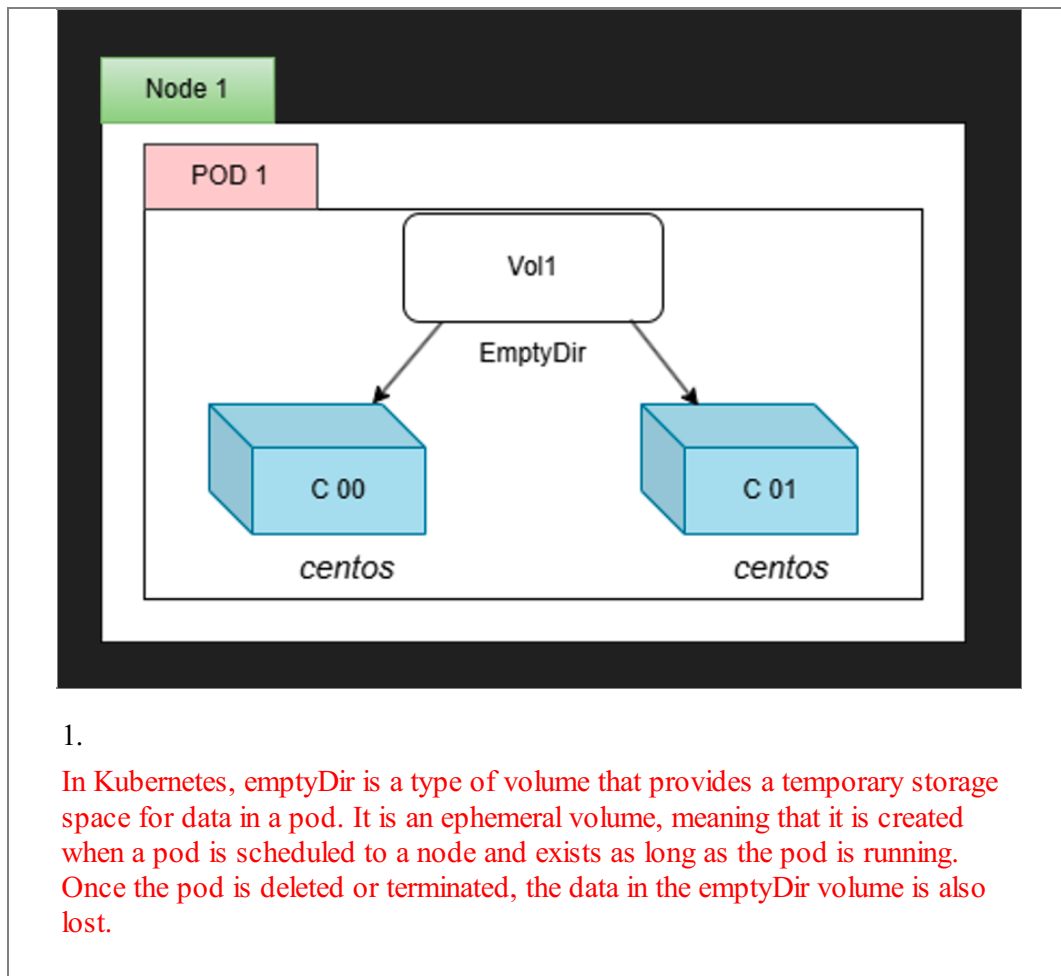

Volumes in Kubernetes

Rohit Surwade

EmptyDir



Manifest to create a pod with EmptyDir volume

```

root@ip-172-31-47-35: /home/ubuntu
apiVersion: v1
kind: Pod
metadata:
  name: myvolemptydir
spec:
  containers:
    - name: c00
      image: centos
      command: ["/bin/bash", "-c", "sleep 15000"]
      volumeMounts: # Mount definition inside the co
ntainer
        - name: voll
          mountPath: "/tmp/xchange"
    - name: c01
      image: centos
      command: ["/bin/bash", "-c", "sleep 10000"]
      volumeMounts:
        - name: voll
          mountPath: "/tmp/data"
  volumes:
    - name: voll
      emptyDir: {}

```

-- INSERT --

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2.

(A Pod named myvolemptydir is defined with two containers (c00 and c01). Both containers use the CentOS image and are configured to run for a long time with the sleep command.

The containers share an emptyDir volume (voll), which is mounted at different paths (/tmp/xchange and /tmp/data) inside the containers.

emptyDir provides a temporary, shared storage space that exists only as long as the Pod is running.

Both containers are able to access the same emptyDir volume, making it easy for them to share files.)

```

apiVersion: v1
kind: Pod
metadata:
  name: myvolemptydir
spec:
  containers:
    - name: c00
      image: centos
      command: ["/bin/bash", "-c", "sleep 15000"]
      volumeMounts: # Mount definition inside the container
        - name: voll
          mountPath: "/tmp/xchange"
    - name: c01
      image: centos
      command: ["/bin/bash", "-c", "sleep 10000"]
      volumeMounts:
        - name: voll
          mountPath: "/tmp/data"
  volumes:
    - name: voll
      emptyDir: {}

```

How to check if Volume is created(c00)

```
root@ip-172-31-47-35: /home/ubuntu# A emptydir.yml
root@ip-172-31-47-35: /home/ubuntu# kubectl apply -f emptydir.yml
pod/myvolemptydir created
root@ip-172-31-47-35: /home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
myvolemptydir 0/2     ContainerCreating 0           8s
root@ip-172-31-47-35: /home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
myvolemptydir 0/2     ContainerCreating 0          12s
B root@ip-172-31-47-35: /home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
myvolemptydir 2/2     Running   0           30s C
root@ip-172-31-47-35: /home/ubuntu# kubectl exec myvolemptydir -c c00 -it -- /bin/bash
[root@myvolemptydir /]# ls
bin  etc  lib  lost+found  mnt  proc  run  srv  tmp  var
dev  home  lib64  media  opt  root  sbin  sys  usr
[root@myvolemptydir /]# cd tmp
[root@myvolemptydir tmp]# ls
ks-script-4luisyla  ks-script-o23i7rc2  ks-script-x6ei4wuw D
[root@myvolemptydir tmp]# cd xchange
[root@myvolemptydir xchange]# ls
[root@myvolemptydir xchange]#
```

3.

commands:

1. `kubectl exec pod-name -c container-name -it -- /bin/bash`
2. `cd tmp/xchange(specified dir as per the manifest)`

As we defined in the manifest the volume is present in the xchnage directory

Creating a file in the Volume

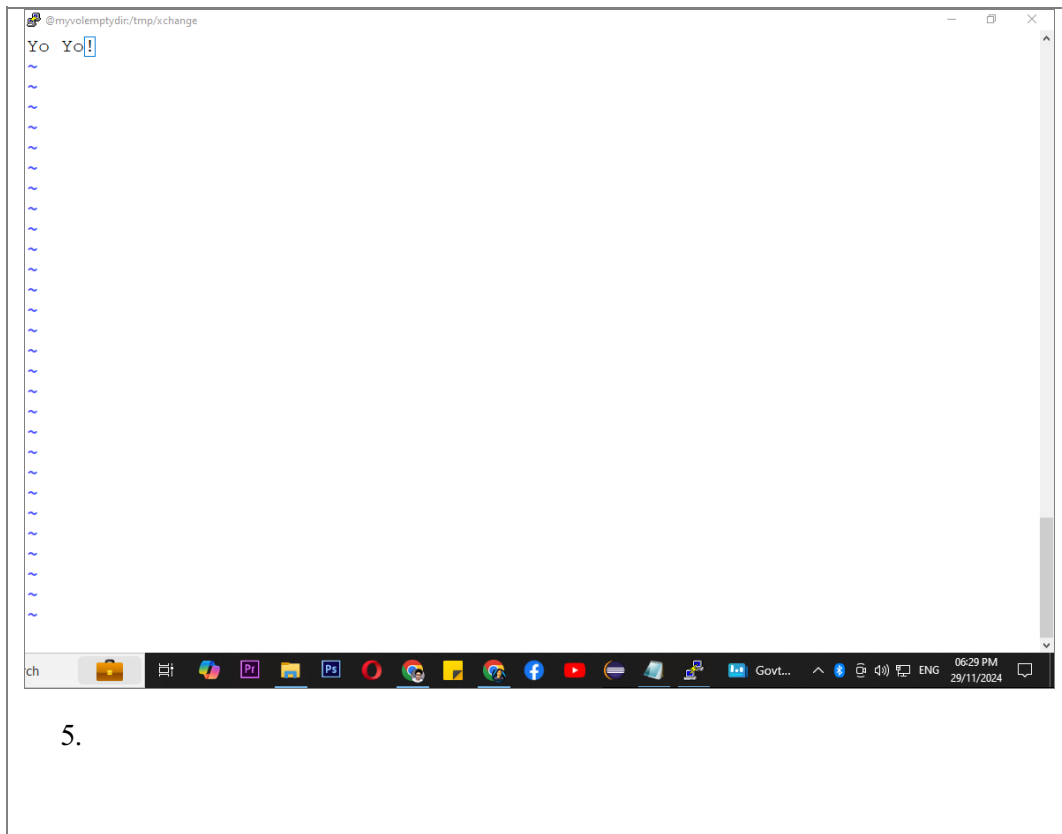
```
@myvolemptydir/tmp/xchange
[root@myvolemptydir xchange]# vi abc.txt
[root@myvolemptydir xchange]#
```



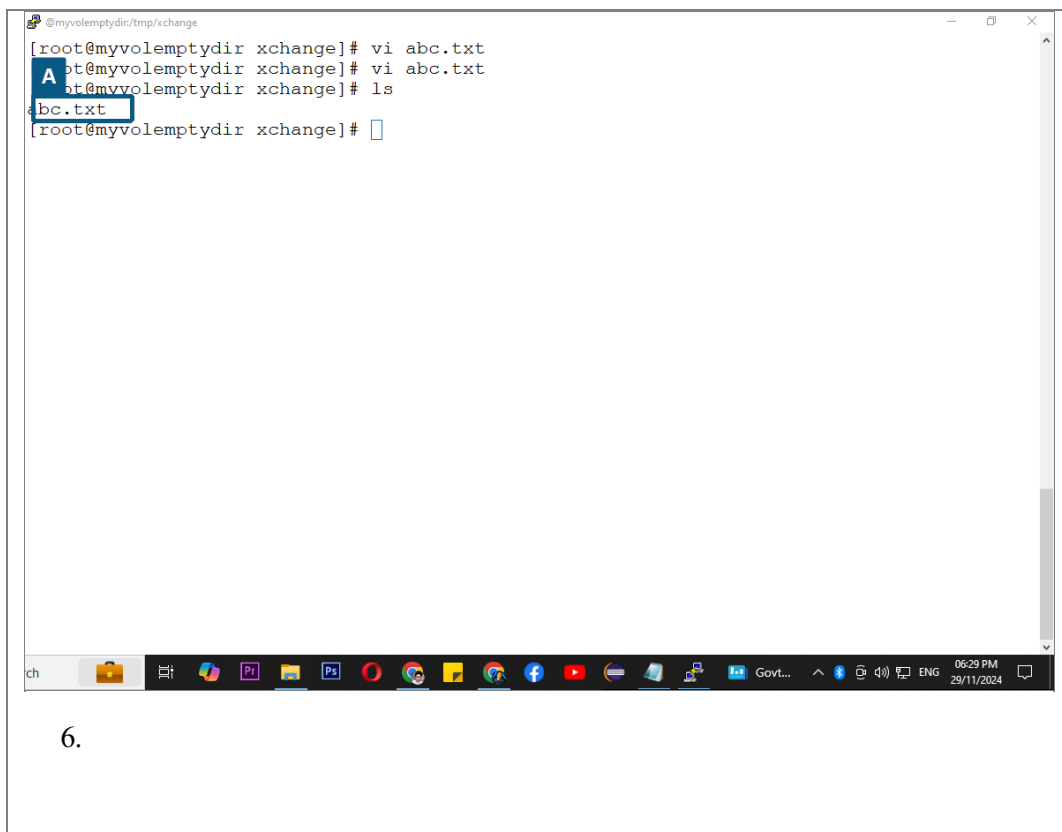
4.

`vi abc.txt`

c00



We have created a file "abc.txt" in the volume from container "c00"



Lets create some files in the volume from container "c01"

```
@myvolemptydir/tmp/data
[root@myvolemptydir xchange]# vi abc.txt
[root@myvolemptydir xchange]# vi abc.txt
[root@myvolemptydir xchange]# ls
abc.txt
[root@myvolemptydir xchange]# exit
exit
root@ip-172-31-47-35:/home/ubuntu# kubectl exec myvolemptydir -c c01 -it -- /bin/bash
[root@myvolemptydir /]# cd tmp
[root@myvolemptydir tmp]# ls
data ks-script-4luisyla ks-script-o23i7rc2 ks-script-x6ei4wu
[root@myvolemptydir tmp]# cd data
[root@myvolemptydir data]# ls
abc.txt
[root@myvolemptydir data]# cat abc.txt
Yo Yo!
[root@myvolemptydir data]# A abc.txt
B [touch brown-rang bring-me-back]
[root@myvolemptydir data]# ls
abc.txt bring-me-back brown-rang
[root@myvolemptydir data]#
```

9.

Now lets switch to container "c00" to see if everything is working as expected

```
@myvolemptydir/tmp/xchange
[root@myvolemptydir data]# exit
exit
root@ip-172-31-47-35:/home/ubuntu# A [kubectl exec myvolemptydir -c c00 -it -- /bin/bash]
[root@myvolemptydir /]# cd tmp/xchange/
[root@myvolemptydir xchange]# ls
abc.txt bring-me-back brown-rang
[root@myvolemptydir xchange]# cat abc.txt
Yo Yo!
Honey singhhhhh!!!
[root@myvolemptydir xchange]#
```

10.

WE CAN SEE ALL THE DATA IS REFLECTING IN CONTAINER c00 AS WELL.